Fiscal Year:	FY 2004	**Task Last Updated:**	FY 09/09/2004
PI Name:	Zhao, Yongliang Ph.D.		
Project Title:	Functional Role Of The Betaig-H3 Gene In High-Energy Heavy Ions-Induced Carcinogenesis		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	10032	**Congressional District:**	15
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2002 Space Biology 02-OBPR-02
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No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA ARC
Contact Monitor:		**Contact Phone:**	
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COI Name (Institution):	Shao, Genze (Columbia University)		
Grant/Contract No.:	NAG2-1637		
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Task Description:	Understanding the cellular and molecular mechanism of radiation carcinogenesis induced by high-energy heavy ions is important for both space radiation protection as well as design of interventional approaches. Our previous data demonstrate that loss of expression of Betaig-h3 gene is causally linked to tumorigenic process in high-LET, alpha particle-irradiated human papillomavirus-immortalized human bronchial epithelial (BEP2D) cells. Furthermore, loss of Betaig-h3 expression is common in many human tumor cell lines and reintroduction of this gene into human lung tumor cells completely abrogates their tumorigenicity, suggesting that Betaig-h3 gene may function as a tumor suppressor in a variety of human tumors. The newly established telomerase (hTERT)-immortalized human bronchial epithelial (hTERT-NHBE) cell model that has normal p53 and p16 functions represents an attractive alternative to further ascertain the mechanism involved in radiation carcinogenesis. In this study, we propose to examine the functional role of Betaig-h3 gene in high-energy heavy ions-induced carcinogenesis using the hTERT-NHBE cells and Betaig-h3 knock-out mice. The proposal has 3 main aims: I) To establish the tumorigenic cell model by irradiating hTERT-NHBE		

	cells using beams of charged heavy ions; II) To demonstrate that loss of Betaig-h3 expression is causally related to the malignant transformation process; and III) To generate mouse embryonic stem cells and knock-out mice with disrupted Betaig-h3 gene. Using the Betaig-h3 null mice model, we will address the following questions: (1) whether mice with disruption of this gene predispose to cancer; (2) Is the incidence of transformation induced by heavy ions higher in Betaig-h3 null embryo fibroblast cells than in wild type cells? (3) Does loss of Betaig-h3 gene in embryo fibroblast cells affect radiation sensitivity and apoptosis induced by high-energy heavy ions?
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Cancer induction by space radiation is a major concern for manned space exploration. Among the various particulate components of ionizing radiation in space, high energy (HZE) heavy ions have been of special concern to radiobiologists and NASA administrators mainly because they contribute about 1% of the particle radiation field in frequency whereas they produce a considerable amount of the whole dose-equivalent because of their large energy deposition and high LET. For the acute effects like cell inactivation are of not so direct interest, late effects including genetic changes and cancer formation are more important for risk estimation. However, for this biological end-point there is no direct human evidence available. The research from this funding project will provide the important information ragarding the malignant transformation and molecular alterations induced by heavy ions.
Task Progress:	1.Transformation of primary human bronchial epithelial cells (HBEC) by heavy ions Human hTERT-immortalized HBEC cells were irradiated by a single dose of 60 cGy of 56Fe ions, and have continuously been cultured for over 5 months. Compared with the parental hTERT-SAEC cells, the sequential stages of alterations have been found in the irradiated cells, including: (1)morphological changes such as smaller size and higher saturation density (2)lost contact inhibition growth which is the phenotype of tumorigenic cells. (3)forming colonies in soft agar which is closely correlated with the tumorigenic ability of transformed cells, and often a necessary step for becoming tumorigenic. Tumorigenic potential of the heavy ion irradiated hTERT-SAEC cells is currently evaluated in nude mice. 2. Downregulation of Betaig-h3 gene is involved in the tumorigenic process of human bronchial epithelial cells induced by heavy ion radiation. Using papillomavirus-immortalized human bronchial epithelial (BEP2D) cells, a radiation induced transformation model has been established by a single 60 cGy dose of 56Fe heavy ion radiation. To determine whether Betaig-h3 gene is involved in the 56Fe ion-induced tumorigenic process, the expression levels of Betaig-h3 gene in tumorigenic cell lines and ability of in vivo tumor suppression by re-introducing the Betaig-h3 gene in tumorigenic cells were determined. We found the expression level of this gene is markedly decreased in three tumorigenic cell lines (56FeT1-T3) compared with parental BEP2D cells. Ectopic expression of Betaig-h3 cDNA in the 56FeT2 tumorigenic cells significantly suppressed their tumorigenicity. Our data strongly suggest that loss of Betaig-h3 expression plays a pivotal role in the tumorigenic process induced by 56Fe heavy ion radiation.
Bibliography Type:	Description: (Last Updated: 09/29/2008)
Articles in Peer-reviewed Journals	Zhao, Y.L.; Shao, G.; Piao, C.Q.; Berenguer, J.; Hei, T.K. "Downregulation of Betaig-h3 gene is involved in the tumorigenic process of human bronchial epithelial cells induced by heavy ion radiation." Radiation Research, submitted , Dec-2004
Presentation	Zhao, Y.L.; Piao, C.Q.; Hei, T.K. "Downregulation of Betaig-h3 gene is involved in the tumorigenic process of human bronchial epithelial cells induced by heavy ion radiation." The 3rd International Workshop on Space Radiation and 15th Annual Space Radiation Health Investigators' Workshop, May 16-20, 2004, Port Jefferson, New York. May-2004